

Healthcare Data Analysis Dashboard

Microsoft Excel Project

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Project Type:

Portfolio Project

Tools Used

- Microsoft Excel 2007
- Pivot Tables
- Pivot Charts
- COUNTIF
- VLOOKUP
- Conditional Formatting

Project Objective:

To analyze healthcare data using Microsoft Excel and create an interactive dashboard using Pivot Tables, Charts, COUNTIF, VLOOKUP, and Conditional Formatting to generate meaningful insights.

Date:

June 2026

- **Dataset Preview:**

	A	B	C	D	E	F	G	H	I	J	K	L
1	Patient ID	Patient Name	Age	Gender	Department	Doctor	Admission Date	Discharge Date	Length of Stay	Treatment Cost	Payment Mode	Month
2	P1001	Patient 1	53	Male	Cardiology	Dr. Sharma	22-4-2025	24-4-2025	2	\$ 126.11	Insurance	April
3	P1002	Patient 2	31	Female	Cardiology	Dr. Sharma	23-4-2025	25-4-2025	10	\$ 510.60	Insurance	April
4	P1003	Patient 3	57	Male	Neurology	Dr. Mehta	24-4-2025	26-4-2025	7	\$ 640.50	Cash	April
5	P1004	Patient 4	74	Female	General Medicine	Dr. Khan	25-4-2025	27-4-2025	10	\$ 582.66	Card	April
6	P1005	Patient 5	52	Male	Pediatrics	Dr. Verma	26-4-2025	28-4-2025	9	\$ 318.95	UPI	April
7	P1006	Patient 6	29	Female	Neurology	Dr. Mehta	27-4-2025	29-4-2025	8	\$ 393.81	UPI	April
8	P1007	Patient 7	74	Female	Gynecology	Dr. Gupta	28-4-2025	30-4-2025	3	\$ 267.83	UPI	April
9	P1008	Patient 8	25	Male	Gynecology	Dr. Gupta	29-4-2025	1-5-2025	7	\$ 492.43	Insurance	April
10	P1009	Patient 9	78	Female	Pediatrics	Dr. Verma	30-4-2025	2-5-2025	3	\$ 163.93	Insurance	April
11	P1010	Patient 10	33	Female	Orthopedics	Dr. Singh	1-5-2025	3-5-2025	5	\$ 240.99	Insurance	May
12	P1011	Patient 11	31	Male	Orthopedics	Dr. Singh	2-5-2025	4-5-2025	4	\$ 318.40	Insurance	May
13	P1012	Patient 12	35	Female	Neurology	Dr. Mehta	3-5-2025	5-5-2025	6	\$ 213.40	Card	May
14	P1013	Patient 13	45	Female	Cardiology	Dr. Sharma	4-5-2025	6-5-2025	5	\$ 224.88	UPI	May
15	P1014	Patient 14	69	Female	Neurology	Dr. Mehta	5-5-2025	7-5-2025	5	\$ 256.52	Insurance	May

Explanation: The dataset includes patient information, hospital details, treatment cost, and admission records used for healthcare data analysis in Microsoft Excel.

- **Data Cleaning:**

	G	H	I
	Admission Date	Discharge Date	
1a	22-04-2025	24-04-2025	
1a	05-01-2025	15-01-2025	
a	10-12-2025	17-12-2025	
i	06-01-2025	16-01-2025	
a	16-07-2025	25-07-2025	
a	07-08-2025	15-08-2025	
a	02-03-2025	05-03-2025	
a	10-04-2025	17-04-2025	
a	24-04-2025	27-04-2025	
i	06-10-2025	11-10-2025	
i	26-10-2025	30-10-2025	
a	30-04-2025	06-05-2025	
1a	03-05-2025	08-05-2025	
a	12-04-2025	17-04-2025	
i	14-06-2025	16-06-2025	
a	14-11-2025	21-11-2025	
a	29-03-2025	07-04-2025	
i	09-11-2025	13-11-2025	
a	12-11-2025	20-11-2025	
1a	22-11-2025	02-12-2025	
i	02-11-2025	03-11-2025	
1a	23-09-2025	30-09-2025	
a	25-07-2025	04-08-2025	
i	01-12-2025	03-12-2025	

	J
y	Treatment Cost
	₹ 11,900.00
	₹ 48,180.00
	₹ 60,438.00
	₹ 54,980.00
	₹ 30,096.00
	₹ 37,160.00
	₹ 25,272.00
	₹ 46,466.00
	₹ 15,468.00
	₹ 22,740.00
	₹ 30,044.00
	₹ 20,136.00
	₹ 21,220.00
	₹ 24,205.00
	₹ 15,082.00
	₹ 42,364.00
	₹ 25,470.00
	₹ 26,828.00
	₹ 40,616.00
	₹ 72,650.00
	₹ 2,691.00
	₹ 20,237.00
	₹ 50,100.00
	₹ 5,932.00

Explanation: The dataset was reviewed for duplicate and missing values. No duplicate records or blank values were found. Date columns were formatted correctly, and the Treatment Cost column was converted from Indian Rupees (₹) to US Dollars (\$) for consistent analysis.

- **Data Formatting:**

	A	B	C	D	E	F	G	H	I	J	K	L
1	Patient ID	Patient Name	Age	Gender	Department	Doctor	Admission Date	Discharge Date	Length of Stay	Treatment Cost	Payment Mode	Month
2	P1001	Patient 1	53	Male	Cardiology	Dr. Sharma	22-04-2025	24-04-2025	2	₹11,900.00	Insurance	April
3	P1002	Patient 2	31	Female	Cardiology	Dr. Sharma	05-01-2025	15-01-2025	10	₹48,180.00	Insurance	January
4	P1003	Patient 3	57	Male	Neurology	Dr. Mehta	10-12-2025	17-12-2025	7	₹60,438.00	Cash	December
5	P1004	Patient 4	74	Female	General Medicine	Dr. Khan	06-01-2025	16-01-2025	10	₹54,980.00	Card	January
6	P1005	Patient 5	52	Male	Pediatrics	Dr. Verma	16-07-2025	25-07-2025	9	₹30,096.00	UPI	July
7	P1006	Patient 6	29	Female	Neurology	Dr. Mehta	07-08-2025	15-08-2025	8	₹37,160.00	UPI	August
8	P1007	Patient 7	74	Female	Gynecology	Dr. Gupta	02-03-2025	05-03-2025	3	₹25,272.00	UPI	March
9	P1008	Patient 8	25	Male	Gynecology	Dr. Gupta	10-04-2025	17-04-2025	7	₹46,466.00	Insurance	April
10	P1009	Patient 9	78	Female	Pediatrics	Dr. Verma	24-04-2025	27-04-2025	3	₹15,468.00	Insurance	April
11	P1010	Patient 10	33	Female	Orthopedics	Dr. Singh	06-10-2025	11-10-2025	5	₹22,740.00	Insurance	October
12	P1011	Patient 11	31	Male	Orthopedics	Dr. Singh	26-10-2025	30-10-2025	4	₹30,044.00	Insurance	October
13	P1012	Patient 12	35	Female	Neurology	Dr. Mehta	30-04-2025	06-05-2025	6	₹20,136.00	Card	April
14	P1013	Patient 13	45	Female	Cardiology	Dr. Sharma	03-05-2025	08-05-2025	5	₹21,220.00	UPI	May
15	P1014	Patient 14	69	Female	Neurology	Dr. Mehta	12-04-2025	17-04-2025	5	₹24,205.00	Insurance	April
16	P1015	Patient 15	66	Female	General Medicine	Dr. Khan	14-06-2025	16-06-2025	2	₹15,082.00	Card	June
17	P1016	Patient 16	70	Female	Neurology	Dr. Mehta	14-11-2025	21-11-2025	7	₹42,364.00	UPI	November
18	P1017	Patient 17	34	Male	Neurology	Dr. Mehta	29-03-2025	07-04-2025	9	₹25,470.00	Card	March
19	P1018	Patient 18	19	Female	Orthopedics	Dr. Singh	09-11-2025	13-11-2025	4	₹26,828.00	UPI	November
20	P1019	Patient 19	70	Male	Neurology	Dr. Mehta	12-11-2025	20-11-2025	8	₹40,616.00	Insurance	November
21	P1020	Patient 20	19	Male	Cardiology	Dr. Sharma	22-11-2025	02-12-2025	10	₹72,650.00	Card	November
22	P1021	Patient 21	18	Female	General Medicine	Dr. Khan	02-11-2025	03-11-2025	1	₹2,691.00	Insurance	November
23	P1022	Patient 22	69	Female	Cardiology	Dr. Sharma	23-09-2025	30-09-2025	7	₹20,237.00	Cash	September
24	P1023	Patient 23	78	Male	Pediatrics	Dr. Verma	25-07-2025	04-08-2025	10	₹50,100.00	Card	July
25	P1024	Patient 24	46	Male	General Medicine	Dr. Khan	01-12-2025	03-12-2025	2	₹5,932.00	UPI	December

Explanation:The dataset was formatted to improve readability and presentation. Bold headers, center alignment, borders, column AutoFit, and consistent formatting were applied to make the data clear and easy to analyze.

- **Formulas Used (COUNTIF & VLOOKUP):**

B2					
			=COUNTIF(Patient_Data!E:E,A2)		
	A	B	C	D	E
1	Department	Total Patients			
2	Cardiology	31			
3	Neurology	40			
4	Orthopedics	38			
5	General Medicine	33			
6	Pediatrics	30			
7	Gynecology	28			
8					
9					

Font

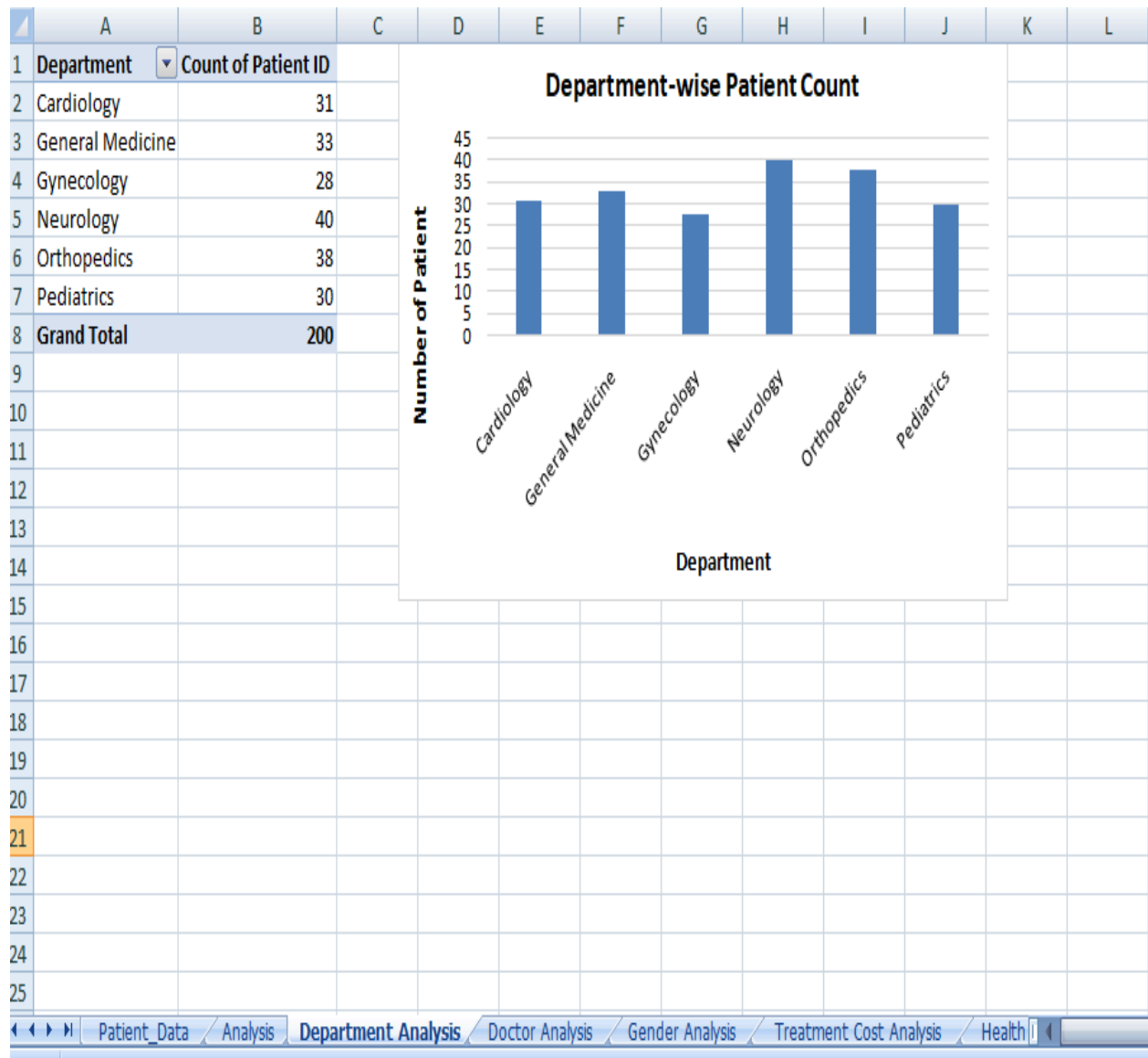
Alignment

=VLOOKUP(A2,Patient_Data!\$E\$2:\$F\$201,2,FALSE)

C	D	E	F	G
	Department	Doctor Name		
	Cardiology	Dr. Sharma		
	Neurology	Dr. Mehta		
	Orthopedics	Dr. Singh		
	General Medicine	Dr. Khan		
	Pediatrics	Dr. Verma		
	Gynecology	Dr. Gupta		

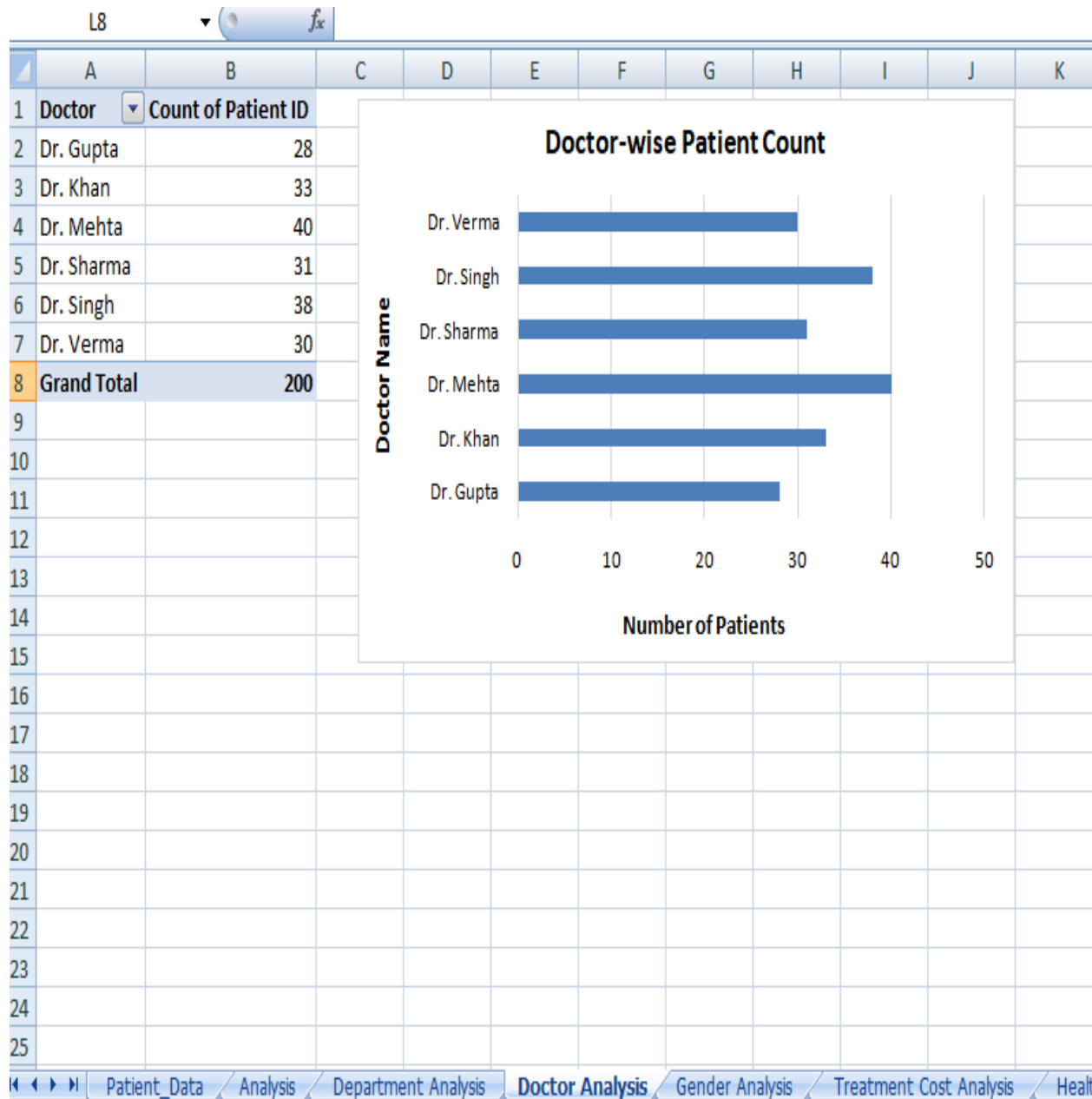
Explanation: COUNTIF and VLOOKUP formulas were used to analyze and retrieve data efficiently. COUNTIF was used to count records based on specific criteria, while VLOOKUP was used to fetch related information from the dataset.

- **Pivot Table 1:**



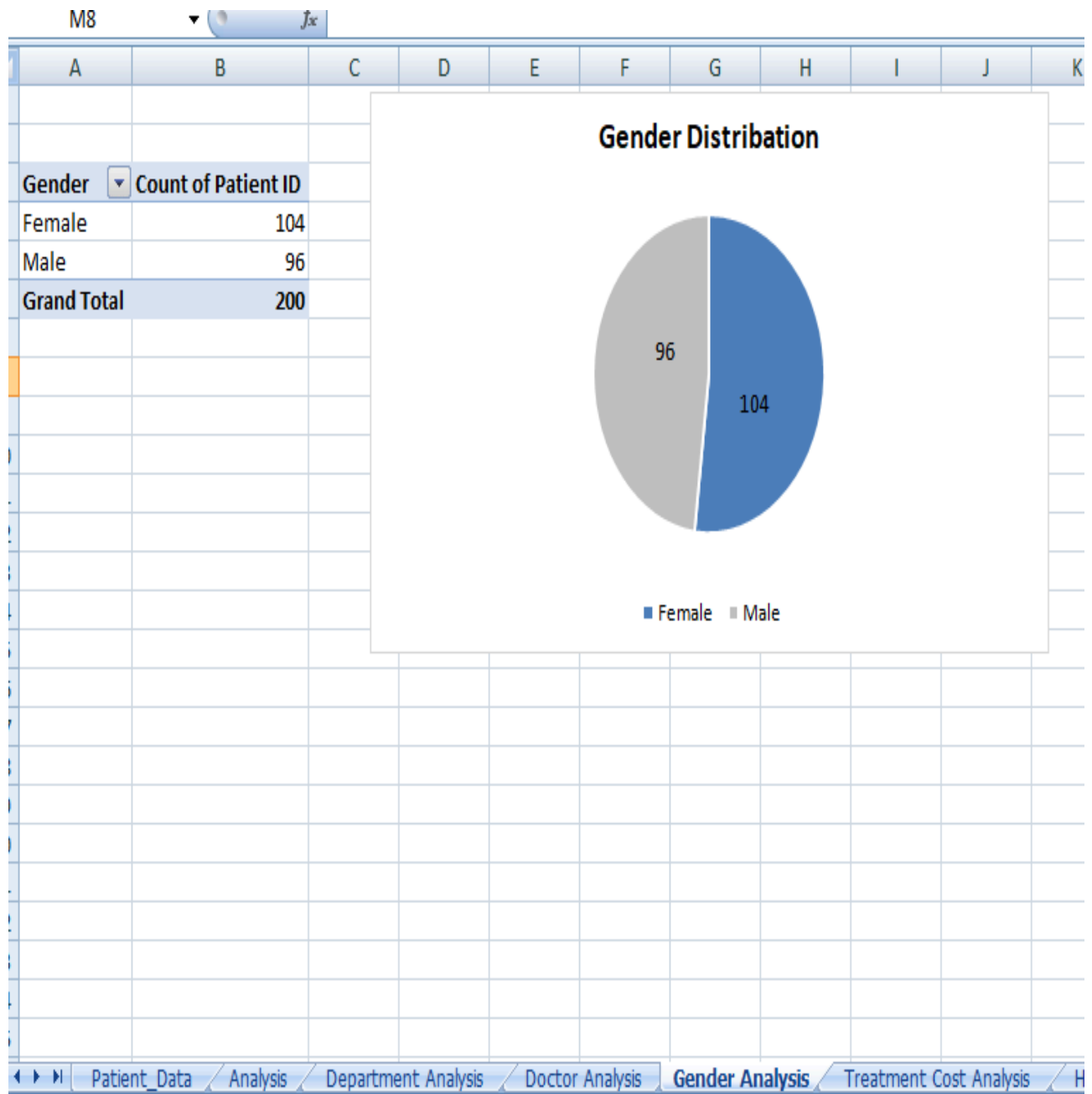
Explanation: This Pivot Table shows the number of patients in each hospital department. It helps identify which department handled the highest and lowest number of patients.

- **Pivot Table 2:**



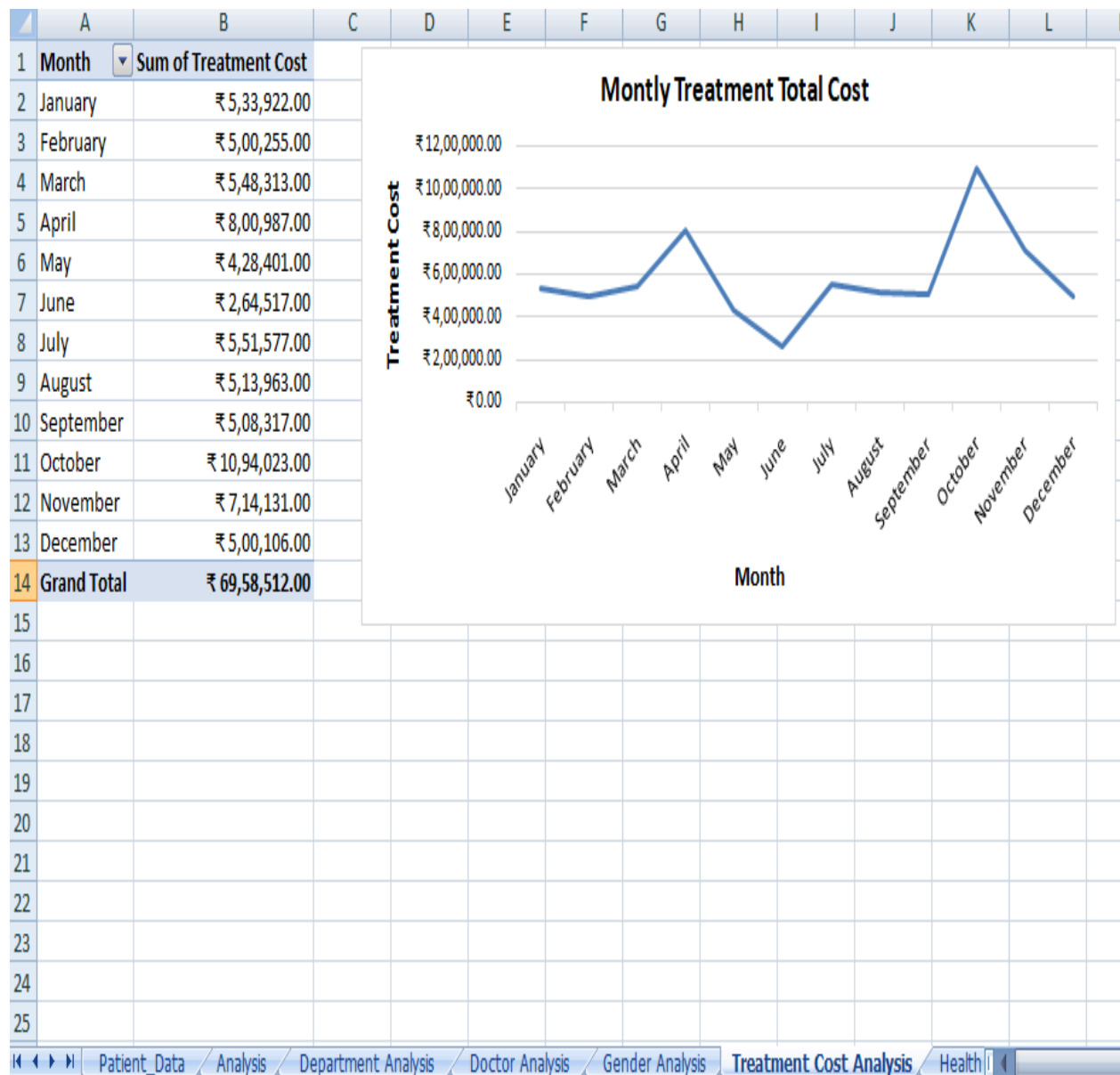
Explanation: This Pivot Table displays the number of patients treated by each doctor. It helps compare the patient distribution among doctors.

- **Pivot Table 3:**



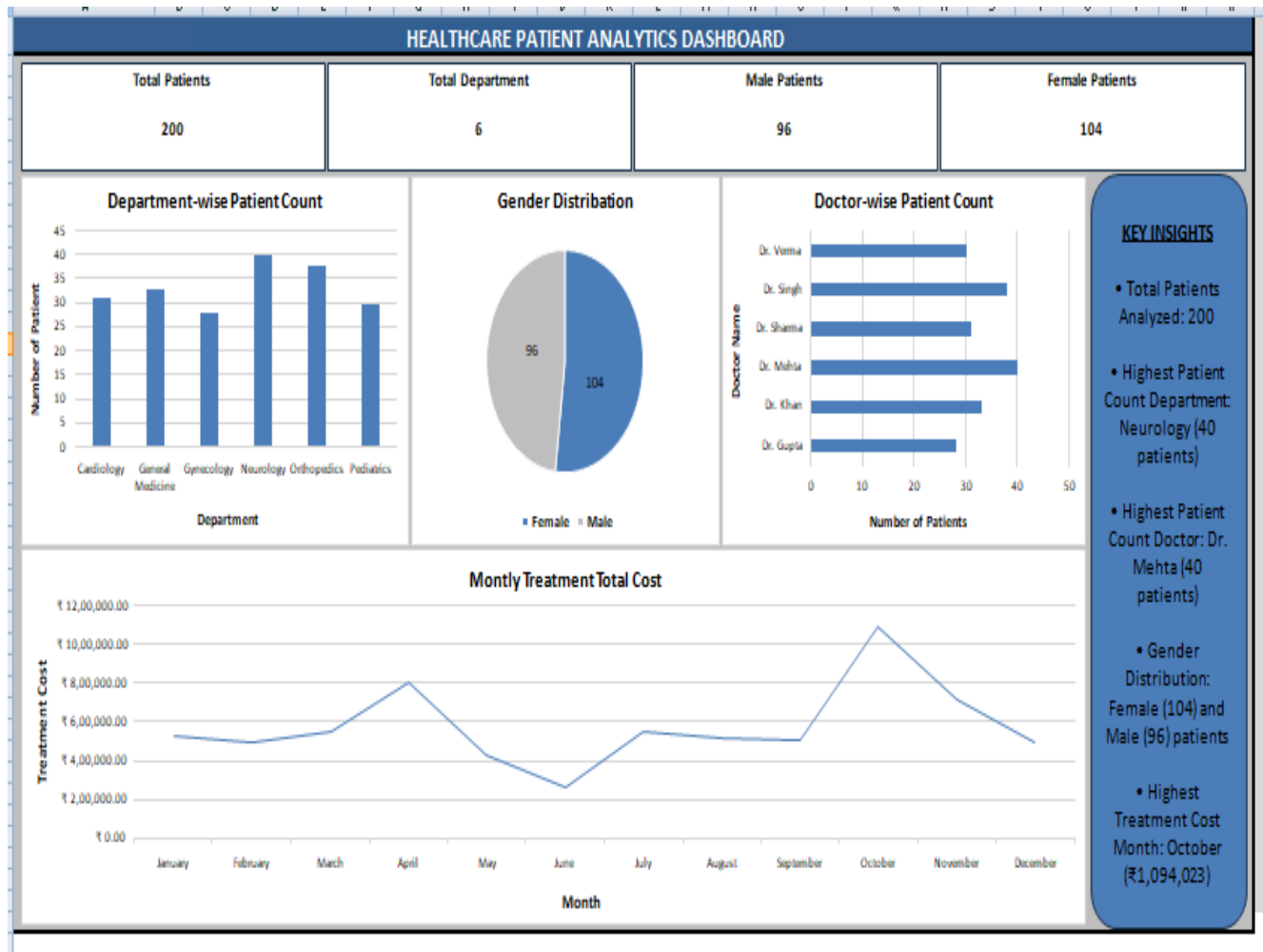
Explanation: This Pivot Table shows the distribution of patients by gender. It provides a quick comparison between male and female patients in the dataset.

- **Pivot table:**



Explanation: This Pivot Table summarizes the total treatment cost for each month. It helps analyze monthly treatment expenses and identify cost trends over time.

- **Final Dashboard:**



Explanation: This interactive dashboard summarizes key healthcare insights using Pivot Tables, Pivot Charts, and Excel formulas. It provides an overview of patient distribution, gender analysis, doctor-wise patient count, department-wise patient count, and monthly treatment costs for effective data analysis.

Key Findings:

- Total patients analyzed: 200
- Neurology recorded the highest patient count (40 patients).
- Dr. Mehta treated the highest number of patients (40 patients).
- Female patients (104) were slightly more than male patients (96).
- October had the highest total treatment cost (₹10,94,023).

Conclusion:

The Healthcare Patient Analytics Dashboard provides a clear overview of patient data, department performance, doctor-wise patient distribution, gender distribution, and monthly treatment costs. The dashboard helps identify important trends and supports data-driven decision-making using Microsoft Excel.